REV A LAYOUT REVIEW. NOT FOR FABRICATION.

2.4 GHz LoRa or FLRC card, same hardware

Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM, FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.

J1
TSW-110-07-G-S
GND 1
SLOT_5V 2
SLOT_5V 3
GND 4
SPI_SCK 5
6 SPI_MOSI
7 SPI_MISO
8 CS_N
9 RESET_N
10 GND

J2
TSW-108-07-G-S
BUSY 1
IRQ 2
TX_EN 3
RX_EN 4
5 AUX0
6
7
8 GND

U1
E28-2G4M27S
RADIO_3V3 1 VCC
GND 2 GND
SPI_MISO_SRC 3 MISO_TX
SPI_MOSI 4 MOSI_RX
SPI_SCK 5 SCK_RTSN
CS_N 6 NSS_CTS
GND 7 GND
RX_EN 8 RX_EN
9 TX_EN
10 GND
11 RESET_N
12 BUSY
13 IRQ
14 AUX0
15
16 GND

U2
AP63203WU-7
RADIO_3V3 1 FB
SLOT_5V 2 EN
SLOT_5V 3 VIN
4 GND
5 SW
6 BST

R5
22
SPI_MISO_SRC 1 — 2 SPI_MISO

ASSEMBLY VARIANT: IPEX ONLY. Selector MUST match manual section 9.2 right image.
MCP23517 controls TX_EN/RX_EN before RF commands. No automatic RX/TX modes.
Buck PCB layout must follow figure 25, 2 oz copper on both layers.

#FLG01
POWER DECLARATION: SLOT_5V
SLOT_5V 1 POWER

#FLG02
POWER DECLARATION: GND
GND 1 POWER

#FLG03
POWER DECLARATION: RADIO_3V3
RADIO_3V3 1 POWER

L1
4.7uH
SW 1 — 2 RADIO_3V3

C1
22uF
SLOT_5V 1 — 2 GND

C2
22uF
RADIO_3V3 1 — 2 GND

C3
22uF
RADIO_3V3 1 — 2 GND

C4
100nF
BST 1 — 2 SW

C5
220uF
SLOT_5V 1 + — 2 GND

C7
22uF
RADIO_3V3 1 — 2 GND

C8
22uF
SLOT_5V 1 — 2 GND

C6
100nF
RADIO_3V3 1 — 2 GND

R1
10k
RADIO_3V3 1 — 2 CS_N

R2
10k
RADIO_3V3 1 — 2 RESET_N

R3
10k
TX_EN 1 — 2 GND

R4
10k
RX_EN 1 — 2 GND

H1
M3 alignment hole

MISO: source -> 22R -> slot A7. Fit OR option: scope/IBIS verify.
Place at module output/socket, before shared bus. No long stubs.

KEYED SLOT: H1 engages the carrier M3 x 25 post before electrical contact.
All headers and passives factory fitted. No user soldering or antenna-selector rework.
Power declarations describe power arriving through the slot and buck output after L1.
Layout authorized; no fabrication release. See DESIGN_REVIEW.md.

TP1
SPI_SCK
SPI_SCK 1 PROBE

TP2
SPI_MOSI
SPI_MOSI 1 PROBE

TP3
SPI_MISO
SPI_MISO 1 PROBE

TP4
CS_N
CS_N 1 PROBE

TP5
RESET_N
RESET_N 1 PROBE

TP6
BUSY
BUSY 1 PROBE

TP7
IRQ
IRQ 1 PROBE

TP8
TX_EN
TX_EN 1 PROBE

TP9
RX_EN
RX_EN 1 PROBE

TP10
IN_5V
SLOT_5V 1 PROBE

TP11
CLIP_GND
GND 1 PROBE

TP12
AUX0
AUX0 1 PROBE

TP13
OUT_3V3
RADIO_3V3 1 PROBE

TP14
REG_GND
GND 1 PROBE

TP15
EN_5V
SLOT_5V 1 PROBE

TP16
SW
SW 1 PROBE

TP17
FB
RADIO_3V3 1 PROBE

TP18
MOD_3V3
RADIO_3V3 1 PROBE

TP19
MOD_GND
GND 1 PROBE

TP20
SW_GND
GND 1 PROBE

SCHEMATIC REVIEW — NOT FOR FABRICATION

Sheet: /
File: e28-2g4m27s-rev-a.kicad_sch

Title: 2.4 GHz LoRa or FLRC card, same hardware

Size: A2	Date: 2026-09-17	**Rev: A-schematic**
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